8. Primary symptomatic lumbar or lumbosacral spinal stenosis (defined as the presence of neurogenic claudication and confirmed by imaging);
9. Diagnosed osteoporosis (T-score of -2.5 or less), spine fragility fracture history, trauma/compression fracture at the intended treatment level, or spinal cancer;
10. Radiographic evidence of any of the following that correlates with predominant physical complaints:
    a. Lumbar/lumbosacral disc extrusion or protrusion >5mm at levels L3-S1;
    b. Lumbar/lumbosacral spondylolisthesis > 2mm at any level;
    c. Lumbar/lumbosacral spondylolysis at levels L3-S1;
    d. Lumbar/lumbosacral facet arthrosis/effusion correlated with facet-mediated pain at levels L3-S1.
11. BMI >40;
12. Advanced generalized systemic disease that limits quality-of-life (QOL) improvements would require a statement of the objective of treatment in such cases;
13. Active, untreated substance abuse disorder.

NOTE: Thermal destruction of the intraosseous BVN must only be performed once per vertebral body from L3-S1 per lifetime. Up to 4 vertebral bodies may be treated during 1 procedure.

Background

Low back pain is a leading cause of disability. Chronic low back pain (CLBP) is a persistent pain in the lumbar region lasting for longer than 3-6 months. Vertebrogenic CLBP is thought to be a distinct type of back pain, originating from degeneration of the vertebral body or vertebral body endplates, resulting in inflammation. The inflammatory response is perceived by the basivertebral nerve (BVN), a sensory nerve that enters the posterior vertebral body and branches out to the superior and inferior endplates. Pain signals are then transmitted to the central nervous system, causing vertebrogenic pain (Fischgrund et al., 2019).

Diagnosis of CLBP and its underlying cause is initiated with a comprehensive history and physical examination. Testing may include bloodwork, standard radiographs, bone scans, computerized axial tomography scans, MRI, electromyography, or a diagnostic injection (NASS, 2022).

CLBP can be treated with physical therapy (e.g., ice, heat, massage, transcutaneous electrical nerve stimulation [TENS], ultrasound), analgesia, anti-inflammatory medications, muscle-relaxing agents, coping skills, minimally invasive injections or procedures (e.g., nerve ablation), surgery, or complementary medicine (e.g., acupuncture, dry needling, magnets, etc.) (NINDS, 2022). Intraosseous BVN ablation using a radiofrequency probe and generator has been suggested as a minimally invasive treatment for CLBP in a subset of patients with specific clinical indications.